Tran The Duong

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M.S. student in Electrical Engineering at KAIST working on multi-object tracking: end-to-end Transformer trackers, probabilistic data association, and identity association for drones, birds, and people. Background in robot control (optimal, risk-sensitive, and learning-based control) with Python, C++, and MATLAB.

Research interests: Deep learning, reinforcement learning, and probabilistic machine learning.

Education

KAIST
M.S. in Electrical Engineering | Daejeon, South Korea Sep. 2024 – Present
Expected graduation: Feb. 2027. Advisor: Prof. Dong Eui Chang.

Hanoi University of Science and Technology
B.S. in Automatic Control | Hanoi, Vietnam Sep. 2019 – May 2024
GPA: **3.89/4.00**. Thesis advisor: Assoc. Prof. Dao Phuong Nam.

Research Experience

Graduate Researcher, CtrlLab, KAIST
Computer vision, robotics, and control; multi-object tracking since Oct. 2025 Sep. 2024 – Present

BeliefCO *Master's thesis, ongoing*
End-to-End Transformer-Based Multi-Object Tracking for Drone and Bird Surveillance

- Built a belief into the CO-MOT tracker: per-track Gaussian motion hypotheses and existence probabilities drive a Poisson multi-Bernoulli association solved by in-network belief propagation.
- Added track dropout, camera-motion compensation, and a query-embedding appearance likelihood.
- Created WOSDETC tracking annotations (birds and drones, CVAT) and recording-level splits; evaluated on BFT, WOSDETC, BSB, and SMOT4SB with every baseline on the same detector.
- Achieved test HOTA of **72.32 on BFT**, **62.91 on WOSDETC**, and **65.77 on BSB**, against 69.72, 59.44, and 63.81 for the strongest baselines, and the highest IDF1 on all four test benchmarks.

Spatio-Temporal Association Learning (STAL) *Python, PyTorch*

- Developed Transformer-based identity association using appearance, geometry, and motion cues; evaluated on DanceTrack, SportsMOT, BFT, and BEE24.
- Doubled inference speed with CUDA graphs at unchanged accuracy; trained with multi-GPU DDP.

Identity Memory for Tracking-by-Detection *Python, PyTorch*

- Developed an online identity memory that plugs into SORT-style trackers and corrects their identity decisions; integrated it into seven trackers, from ByteTrack to TrackTrack, and evaluated on DanceTrack, SportsMOT, MOT17, and PersonPath22 with Optuna-tuned settings.

Robotics and Control Projects

Wheeled-Bipedal Robot Modeling and Control
KAIST | MATLAB/Simulink, CasADi, C++, Gazebo Sep. 2024 – Oct. 2025

- Modeled the robot as a wheeled inverted pendulum and formulated a control-Lyapunov-function MPC in CasADi for velocity tracking under state and torque limits, with polynomial reference trajectories.
- Wrote a C++ Gazebo plugin with Protocol Buffers messages that exchanges joint states, IMU data, and torque commands with Simulink for co-simulation.

Risk-Sensitive Stochastic Control (RS-JiLQG)
KAIST graduate course project (EE899), with a classmate | Python, NumPy Fall 2025

- Derived a risk-sensitive iterative LQG for nonlinear jump-diffusion systems: an entropic-risk objective with Bernoulli-approximated slip jumps and a jump-augmented backward pass.
- Built a jump-diffusion simulator of a wheeled inverted pendulum with EKF/UKF state estimation; compared mean and entropic costs across risk levels and against iLQG and MPC in Monte Carlo rollouts.

Additional Experience

Engineering Intern, Viettel Aerospace Institute
Embedded systems software Sep. 2023 – Feb. 2024

- Designed user interfaces for embedded systems; worked with C++, MATLAB, and Simulink.

Undergraduate Researcher, HUST
Reinforcement learning and control 2022 – 2024

- Investigated output-feedback adaptive dynamic programming (ADP) and reinforcement learning control for a two-wheeled inverted pendulum in the undergraduate thesis.
- Studied ADP-based control for mobile robots with wheel slip and wind turbines with disturbance observers; designed a PID controller for a resistance furnace.

Technical Skills

Programming: Python, C++, C, MATLAB/Simulink.

Deep learning: PyTorch, multi-GPU training (DDP), CUDA-graph inference optimization, Transformer detectors (Deformable DETR), YOLOX.

Multi-object tracking: End-to-end trackers (CO-MOT, MOTIP); tracking-by-detection (ByteTrack, OC-SORT, BoT-SORT, DiffMOT); Poisson multi-Bernoulli filtering; TrackEval metrics (HOTA, IDF1, MOTA) and benchmark test-server submissions.

Robotics and control: ROS 2, PX4, Gazebo, CasADi; optimal, risk-sensitive, and adaptive control; reinforcement learning (adaptive dynamic programming); state estimation (EKF, UKF).

Tools: Linux, Git, Optuna, SLURM (KISTI Neuron supercomputer), CVAT, LaTeX.

Languages: Vietnamese (native), English.

Scholarships and Awards

- **KAIST Scholarship**, M.S. program.
- **Best Graduation Thesis**, HUST thesis defense council, Feb. 2024.
- **Academic excellence scholarships**, HUST, six semesters (2020–2023).
- **Corporate scholarship for outstanding students**, 2023.

Additional Projects

PX4 Offboard Position Control with ROS 2
Python, ROS 2, PX4 Nov. 2024

- Implemented a ROS 2 node that sends position commands to PX4 and monitors vehicle state and command acknowledgments.

Pattern Recognition with RBM Features and Neural Classification
Python, NumPy, TensorFlow | KAIST coursework Dec. 2024
Implemented a restricted Boltzmann machine feature extractor and neural classifier for MNIST, with sampling, normalization, and separate training, validation, and test sets.